

Aim: I1Extends I2 and I3(I-Interface)

Steps for Execution:

1. Create a folder named assignment.
2. Create a Java file inside it:
`I1extends I2 and I3.java`
3. Write/save the program in the file.
4. Open the Eclipse IDE.
5. Compile the program.
6. Run the program.
7. Output:

I1 method

I2 method

I3 method

Algorithm:

1. Start.
2. Create interface **I1** with the diff() method.
3. Create interface **I2** with the sum() method.
4. Create interface **I3** that extends both **I1 and I2**.
5. Add the add() method to I3.
6. Create class **Demo** that implements I3.
7. Implement diff(), sum(), and add() methods in Demo.
8. Create an object d of Demo class.
9. Call diff().
10. Call sum().
11. Call add().
12. Display the output.
13. Stop.

Program:

```
package corejava;
```

```
//I1 is an interface
```

```
interface I1 {
```

```

// Method declared for I1
void diff();
}

// I2 is another interface
interface I2 {

// Method declared for I2
void sum();
}

// I3 extends both I1 and I2

// Therefore, I3 inherits diff() from I1 and sum() from I2
interface I3 extends I1, I2 {

// Method declared for I3
void add();
}

// Demo class implements I3

// So Demo must implement diff(), sum(), and add()
class Demo implements I3 {

@Override

// Implementation of add() method from I3
public void add() {
    System.out.println("I3 method");
}

@Override

// Implementation of sum() method from I2
public void sum() {
    System.out.println("I2 method");
}

@Override

// Implementation of diff() method from I1
public void diff() {
    System.out.println("I1 method");
}
}

```

```
}  
}  
  
// Main class  
public class I1ExtendsI2I3 {  
    public static void main(String[] args) {  
        // Creating an object of Demo class  
        Demo d = new Demo();  
        // Calling diff() method  
        d.diff();  
        // Calling sum() method  
        d.sum();  
        // Calling add() method  
        d.add();  
    }  
}
```

Output:

I1 method
I2 method
I3 method

Output Screenshot:

```
Console X
<terminated> !!ExtendUI213 (!) [Java Application] C:\Program Files\Java\jdk-25.0.3\bin\javaw.exe (29-Aug-2025, 8:14:19 pm - 8:14:20 pm elapsed 0:00:00) (pid: 28756)
T1 method
T2 method
T3 method
}
```